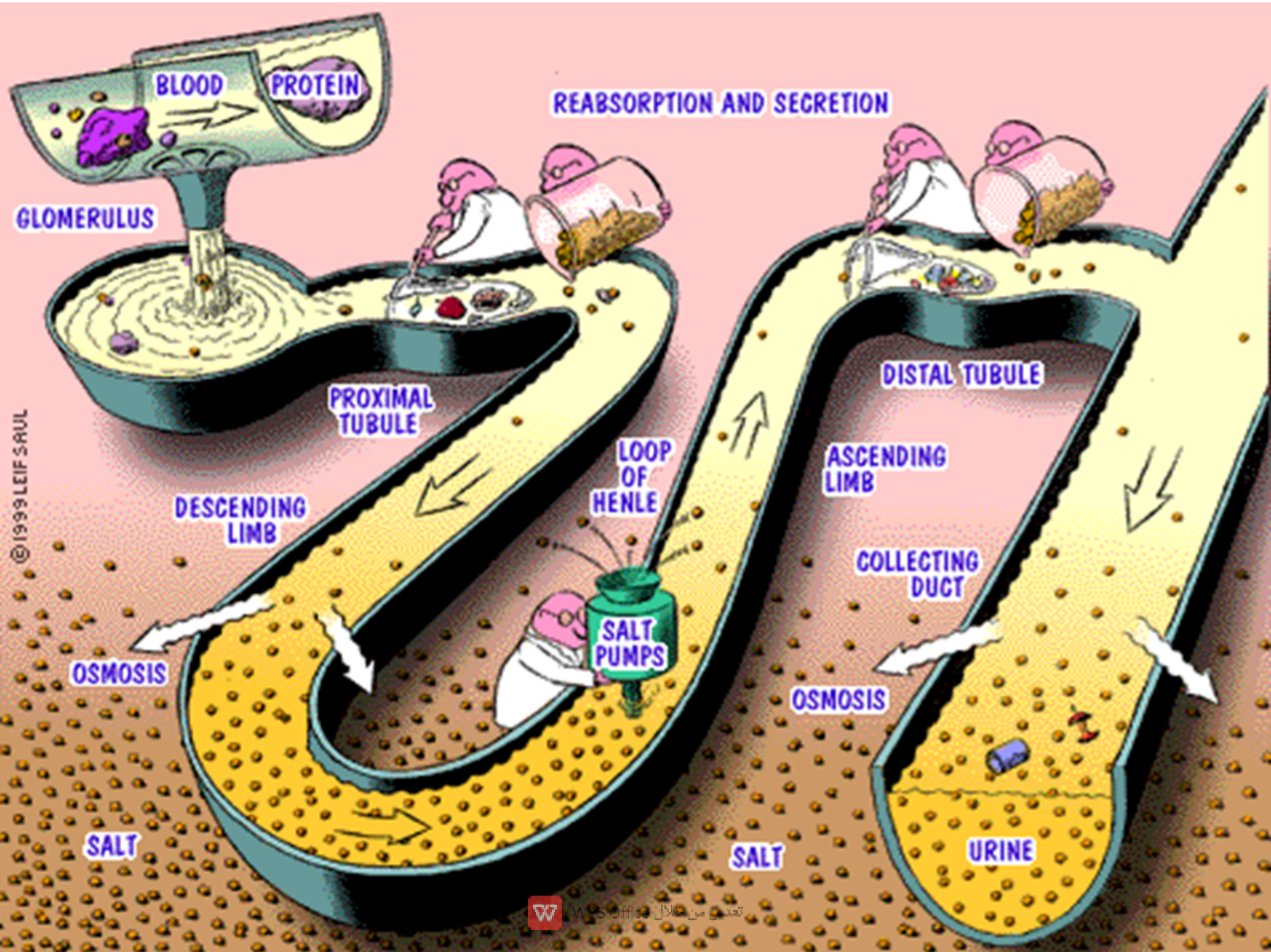




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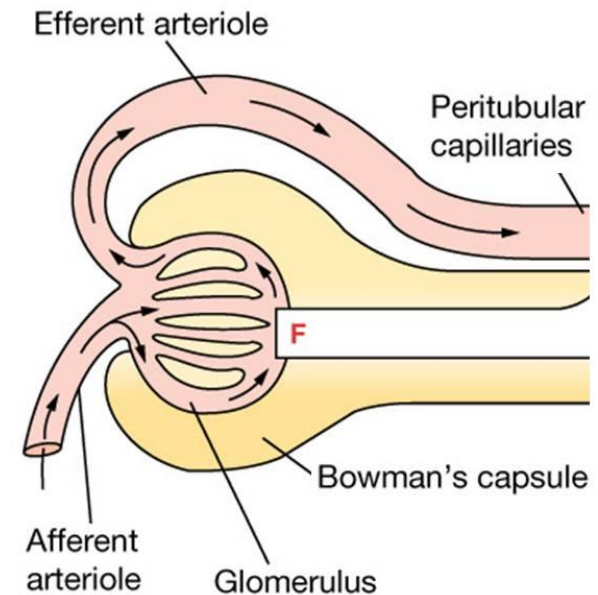
Glomerular Function GFR

Dr. Wafa Mubarak

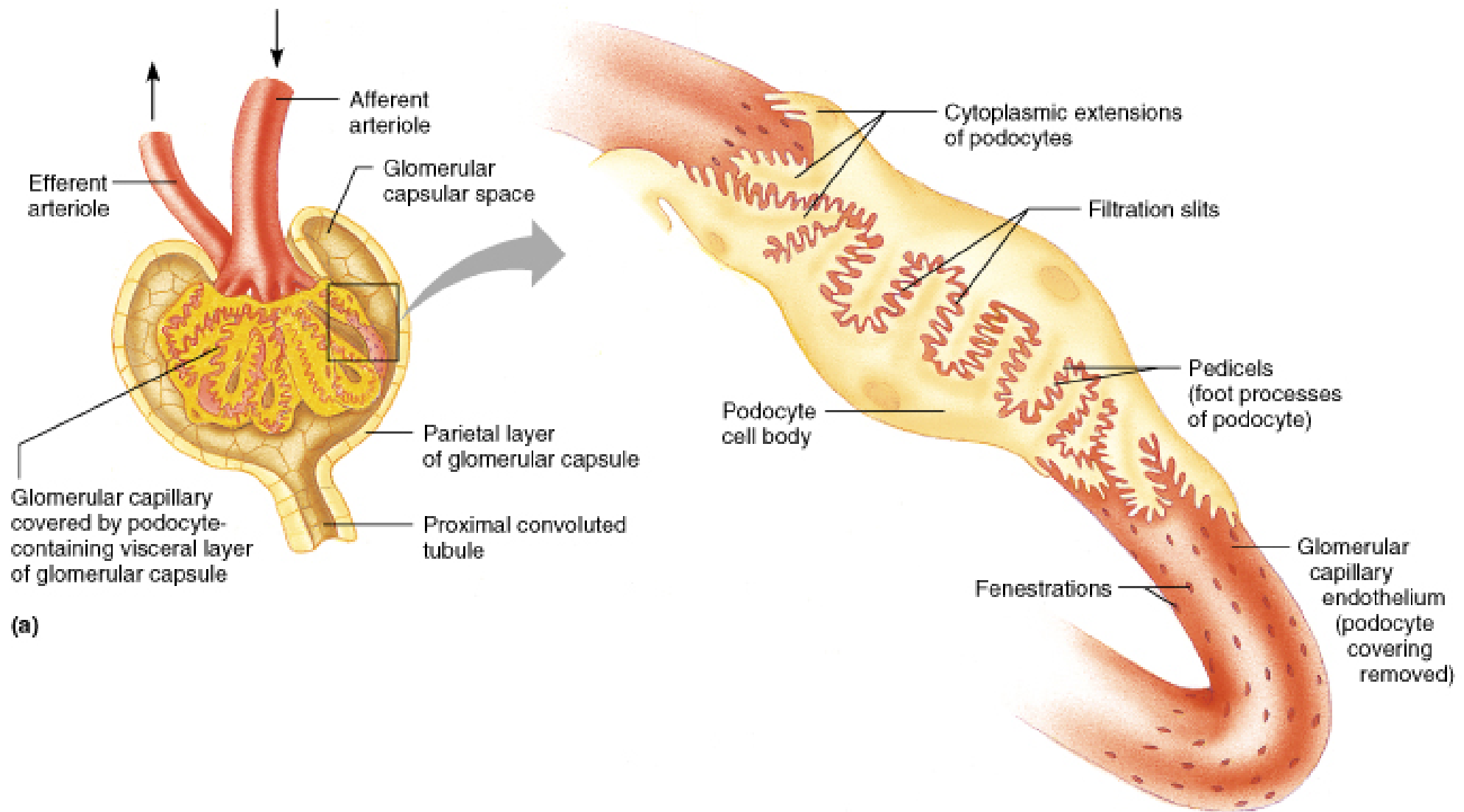


1- Glomerular filtration

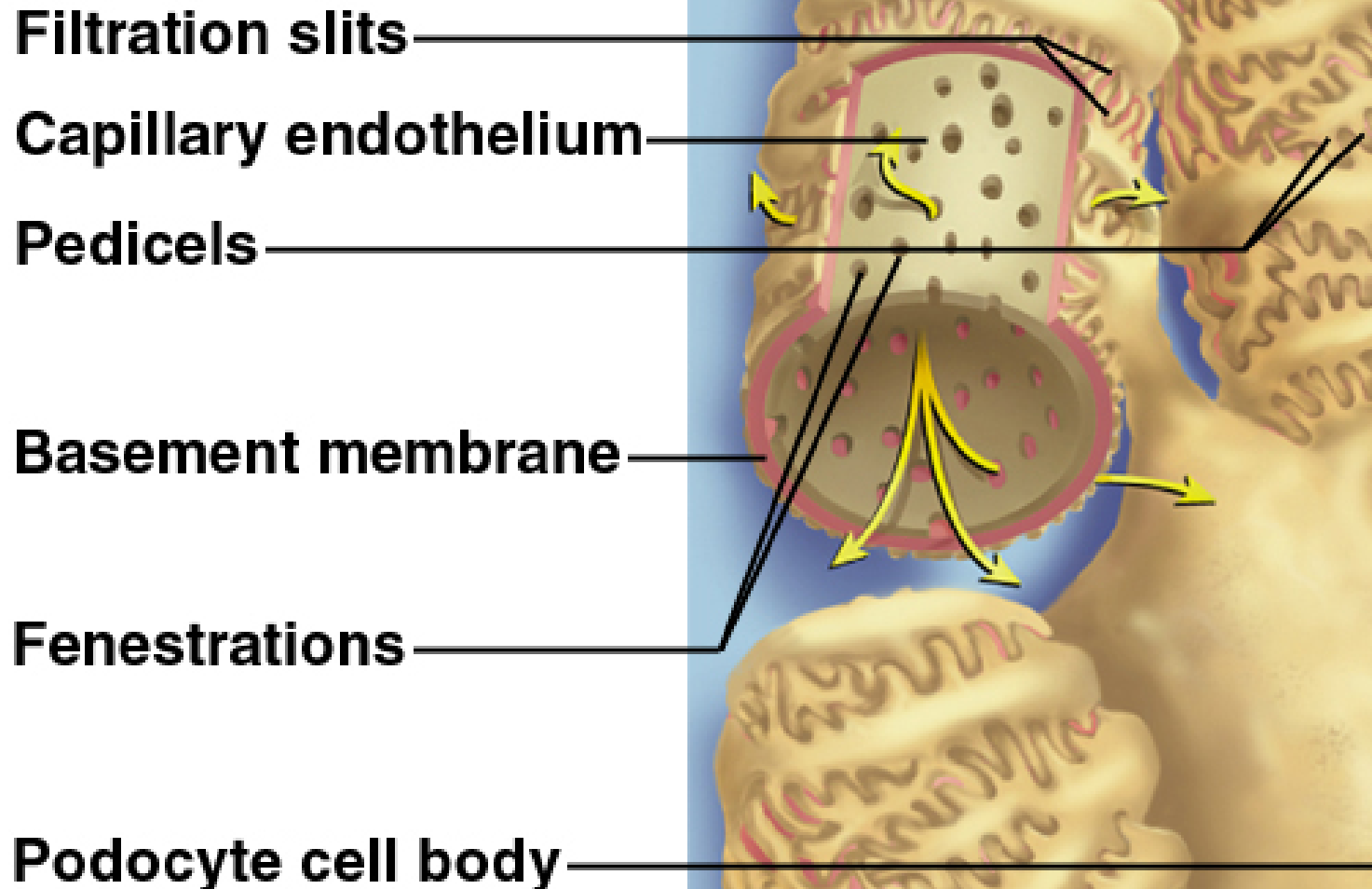
- Is the passage of fluid and crystalloid from glomerular capillary to the bowman's capsule through the (triple) filtration membrane.
- the *triple filtration membrane* –
 - It is composed of three layers
 1. Fenestrated endothelium of the glomerular capillaries
 2. Basement membrane composed of fused basal laminas of the other layers
 3. membrane of the glomerular capsule



Triple Filtration Membrane



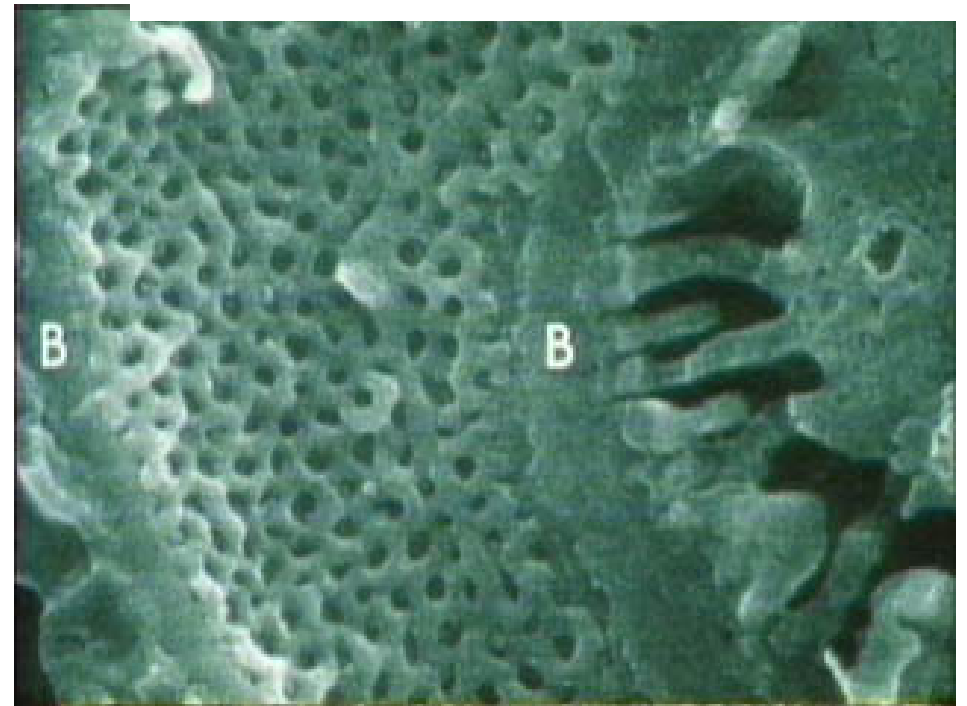
Glomerulus



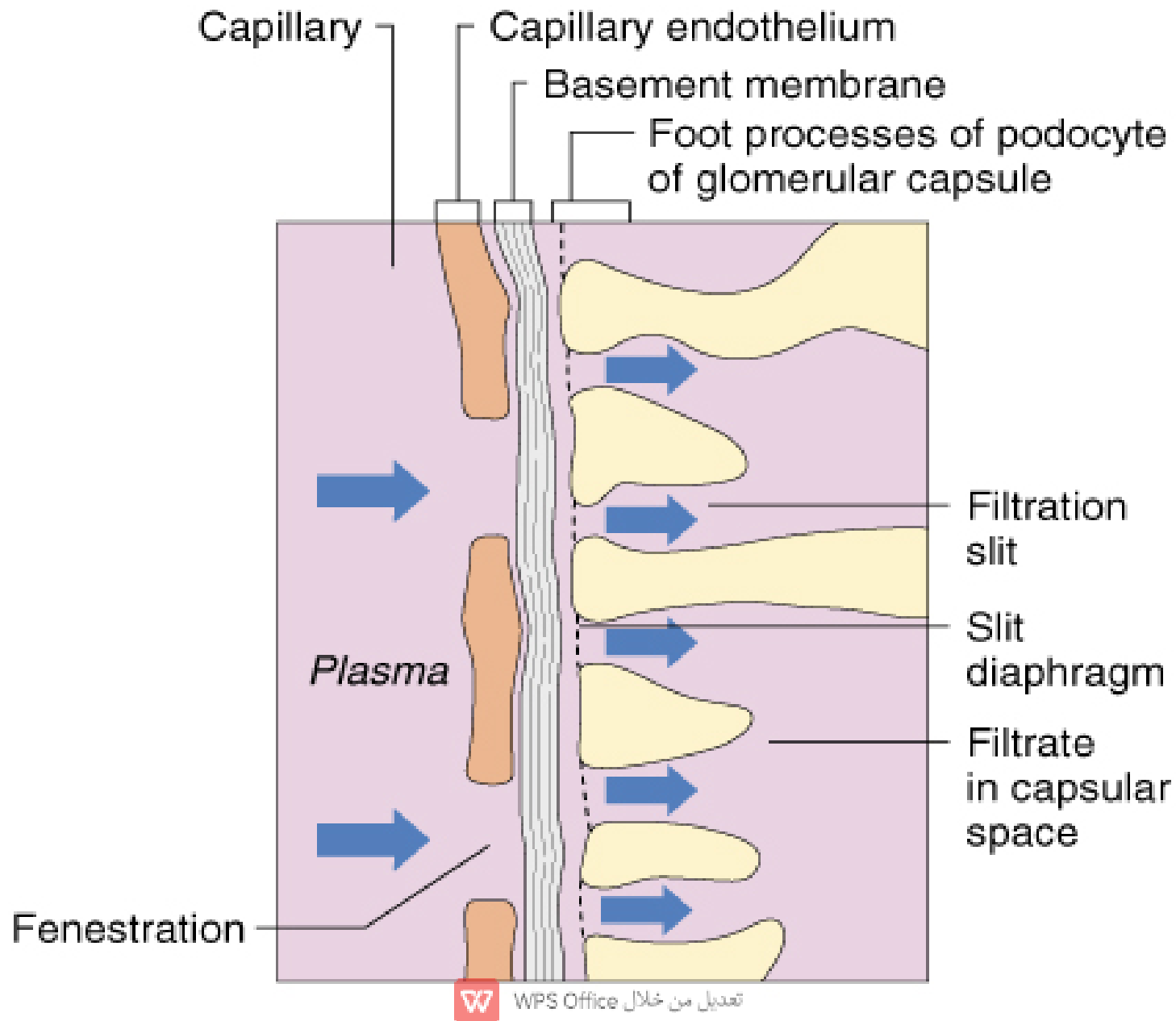


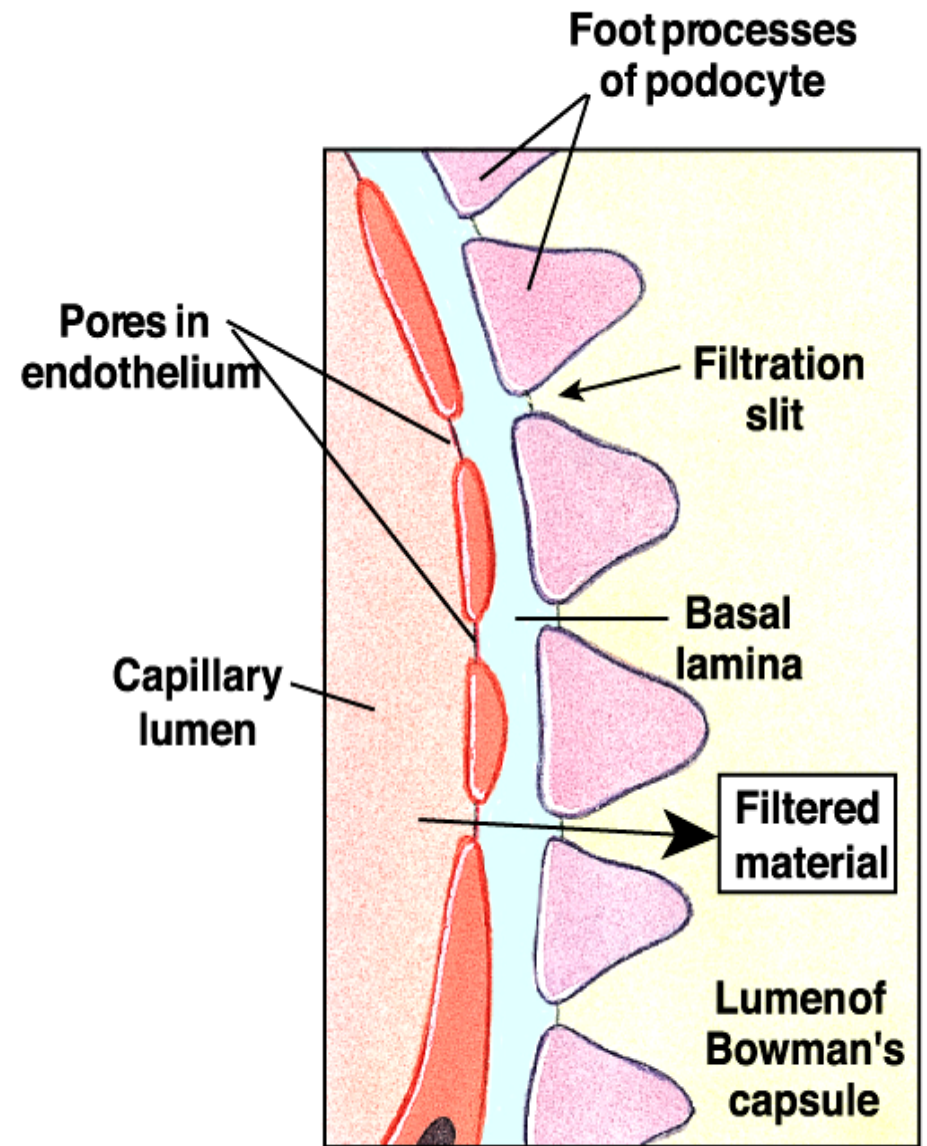
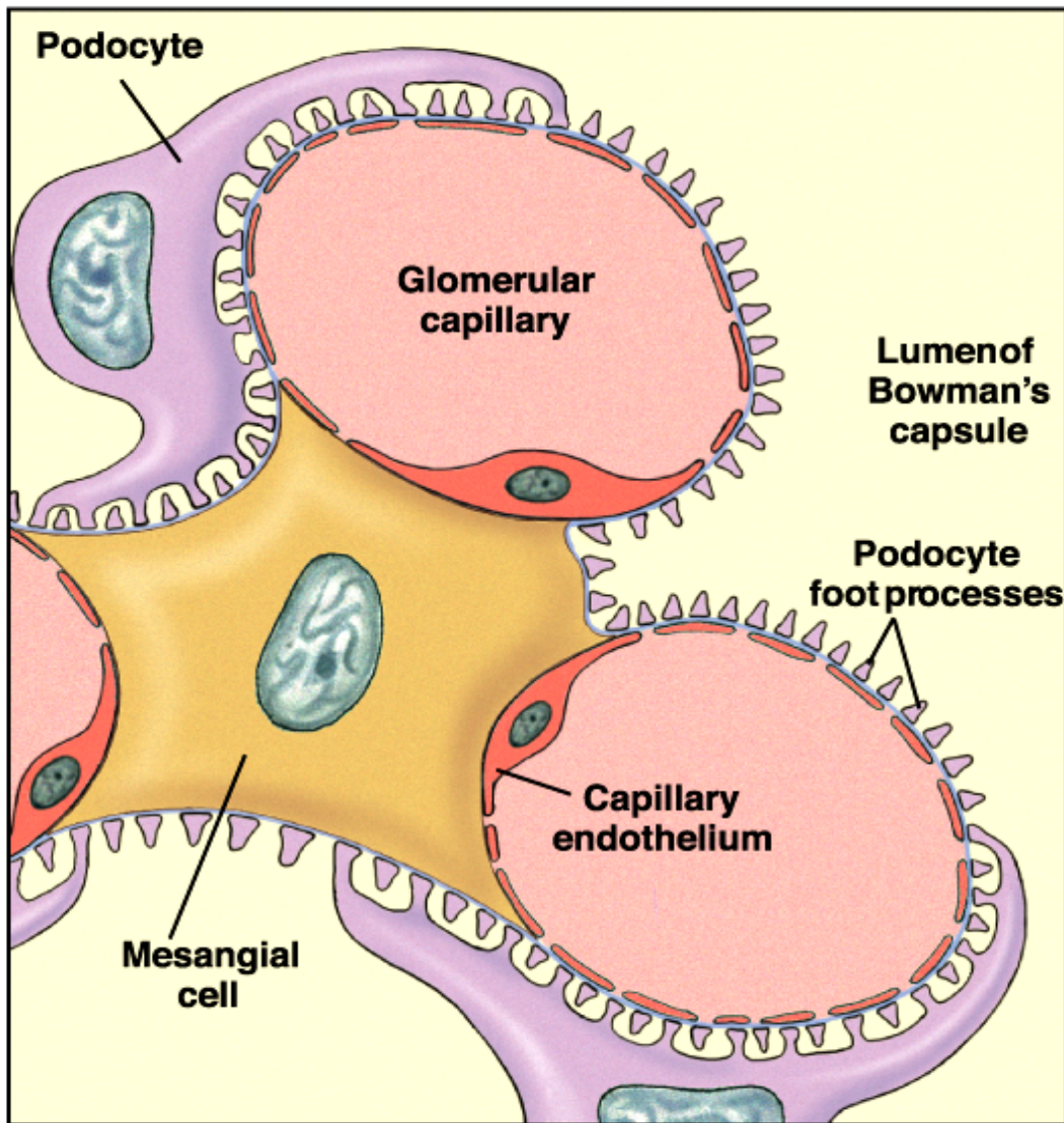
SEM pictures showing a podocyte and the structure of the glomerulus. The pedicles wrap around the capillaries to create the fenestrated membrane.

This is the inside of the capillary with a clear picture of the fenestrations. B is the basement membrane.

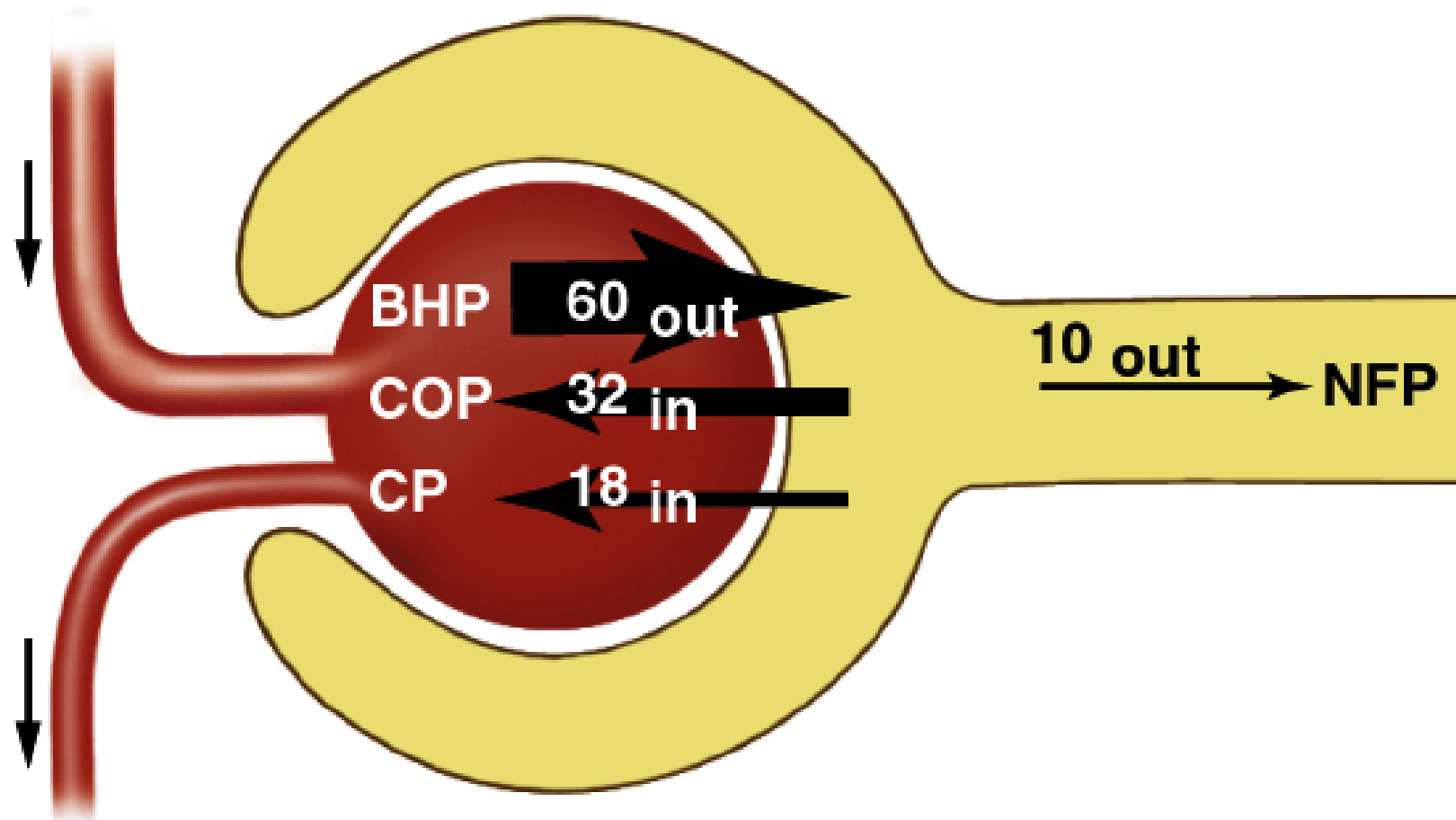


Triple Filtration Membrane





Glomerular Filtration Forces



Blood hydrostatic pressure (BHP)

Colloid osmotic pressure (COP)

Capsular pressure (CP)

Net filtration pressure (NFP)

60 mmHg out

–32 mmHg in

–18 mmHg in

10 mmHg out

Glomerular Filtration Rate (GFR)

- **GFR**

- Is the volume of plasma that is filtrated by all glomeruli in both kidneys (1.3×2 million) per unit of time (ml/min)

- **Value**

- 125 ml/Min = 180 L/Day.

- **Composition**

- Similar to the plasma except it contain less Ca,

- NO RBCs.

- NO plasma protein.

Factors determine the GFR

- The **permeability**
 - Is affected by 4 factors:
 1. Size (less than 8 nm)
 2. Molecular weight (less than 60.000 Dalton)
 3. Charge (+ve charges)
 4. Shape (regular)
- The **surface area**
 - Normally is .8 m².
 - regulated by Mesangial cells
- The **NET filtration pressure** (which include the hydrostatic and oncotic pressure difference)



Factor contract mesengial cells(S A decreased)	Factor relaxes mesengial cells (S A increased)
Angiotensin II Endothelins -Noradrenalin -Thromboxane A 2 - leukoterines - Histamine	A N P Prostaglandins Dopamine



Factor affecting GFR

- Increased the permeability of filtration membrane:
 - - in abnormal condition, such as hypoxia, lack of blood and toxic agent, these lead to increased GFR.
- Change in glomerular HP :-
 - 1/ systemic blood pressure
 - 2/ Resistance in afferent A
 - 3/ Resistance in efferent A



Factor affecting GFR

- Change in capillary oncotic Pressure :-
 - increase in capillary OP as in dehydration decreases GFR
 - decrease in capillary OP as in hypoalbuminemia increase GFR
- Capsular pressure:-
 - Increase in capsular pressure will decreases GFR .
 - » Ureter obstruction, by stones ,tumor and bilharzias.
 - » Edema of the kidney in side tight renal capsule .
- Renal Plasma Flow (RPF) :-
 - increase RPF increase GFR and vice versa

REGULATION OF GFR

- If the GFR is too high:
 - Needed substances cannot be reabsorbed quickly enough and are lost in the urine
- If the GFR is too low:
 - Everything is reabsorbed, including wastes that are normally disposed of



REGULATION OF GFR

- Three mechanisms regulate the GFR
 - Renal autoregulation (intrinsic system)
 - Neural controls
 - Hormonal mechanism (the renin-angiotensin system)



INTRINSIC REGULATION

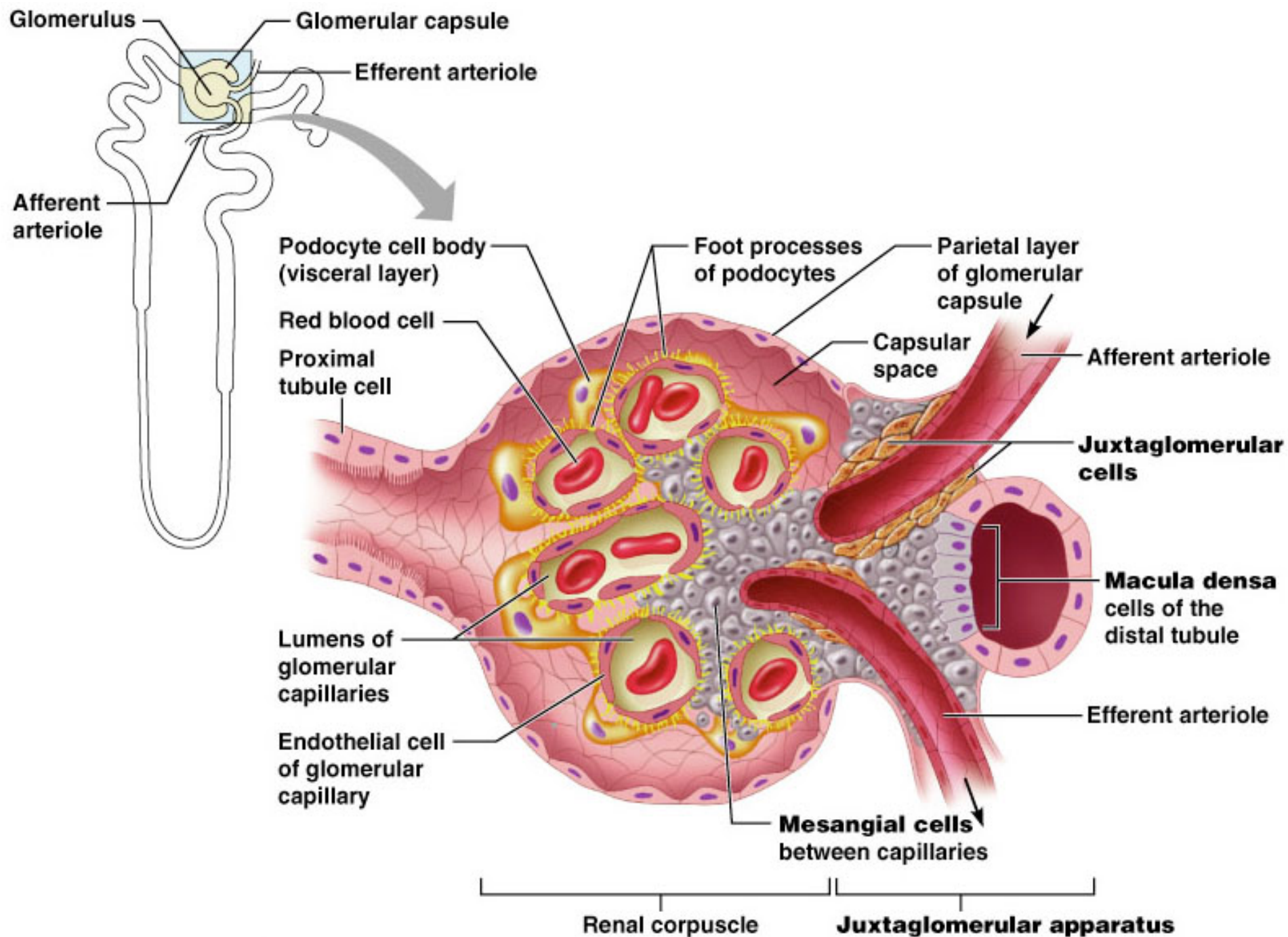
- Under normal conditions, renal autoregulation maintains a nearly constant glomerular filtration rate
- Autoregulation entails two types of control
 - Myogenic – responds to changes in pressure in the renal blood vessels
 - Flow-dependent tubuloglomerular feedback
 - senses changes in the juxtaglomerular apparatus



TUBULAR GLOMERULAR FEED BACK

- THE macula densa response to Na^+ in the filtrate delivered to DT .
- When GFR increase macula densa inhibit renin secretion .
- When GFR decrease macula densa produce vasoconstrictors which lead to vasoconstriction of Afferent Arteriole this raise the capillary Hydrostatic Pressure .





EXTRINSIC REGULATION

- At rest:
 - Renal blood vessels are maximally dilated
 - Autoregulation mechanisms



EXTRINSIC REGULATION

- Under stress:
 - Norepinephrine is released by the sympathetic nervous system
 - Epinephrine is released by the adrenal medulla
 - Afferent arterioles constrict and filtration is inhibited
- The sympathetic nervous system also stimulates the renin-angiotensin mechanism



HORMONAL REGULATION

RENIN ANGIOTENSIN SYSTEM :-

- * Is triggered when the JG cells release renin
 - Renin acts on angiotensinogen to release angiotensin I
 - Angiotensin I is converted to angiotensin II
 - Angiotensin II:
 - Causes mean arterial pressure to rise
 - Stimulates the adrenal cortex to release aldosterone
 - As a result, both systemic and glomerular hydrostatic pressure rise



RENIN- ANGIOTENSIN SYSTEM

- Renin release is triggered by:
 - Reduced stretch of the granular JG cells
 - Stimulation of the JG cells by activated macula densa cells
 - Direct stimulation of the JG cells via $\beta 1$ -adrenergic receptors by renal nerves



other hormones regulate GFR

- Prostaglandin E2 is vasodilator .
- ADH Important for concentration of urine , is cause vasoconstriction.
- serotonin is vasoconstrictor .
- Dopamine is vasodilator .
- ANP is vasodilator.



Renal Clearance

- The volume of plasma that is completely cleared of a particular substance per unit of time
- Renal clearance tests are used to:
 - Determine the GFR
 - Detect glomerular damage
 - Follow the progress of diagnosed renal disease
 - Assess tubular transport mechanism of certain substances .



Renal Clearance

- $RC = UV/P$
- RC = renal clearance rate
- U = concentration (mg/ml) of the substance in urine
- V = flow rate of urine formation (ml/min)
- P = concentration of the same substance in plasma



Characteristic of substance suitable for measuring GFR

- Freely filtered
- Not reabsorbed by tubules
- Not secreted by tubules
- Not metabolized
- Not stored
- Not toxic
- Has no effect on GFR
- Easy to measure in plasma and urine



SUBSTANCES USED FOR MEASURING GFR

- Inulin:- fructose polymer ,molecular weight 5200
- Procedure :- an intravenous load of inulin is administered followed by sustain infusion of inulin until plasma level is constant , then urine is collected for a timed period to get urine flow rate and urinary concentration of inulin
- Inulin is the best substance for measuring GFR because it satisfies criteria .



SUBSTANCES USED FOR MEASURING GFR

- Creatinine :-
 - Product of metabolism of creatine phosphate in muscles
 - It used because its plasma concentration constant and is related to lean body mass .
 - Normal plasma concentration 0.1-1mg/dl
 - Small amount of it secreted in urine there for GFR is higher than that of inulin



- In laboratories plasma and urine level are both elevated ,so the GFR becomes more accurate
- Procedure : urine collecting for 24 hrs
- Sample of plasma to measure creatinine



Estimates of GFR

- Inulin neither secreted or reabsorbed
- Clearance of inulin approximates GFR

$$\text{GFR} = \frac{[U]_{\text{inulin}} V}{[P]_{\text{inulin}}}$$

- Creatine is secreted, so Cr clearance overestimates GFR

Any QUESTION

THANK YOU

